The Giant African Snail (*Lissachatina fulica*): A Comprehensive Review of Biology, Impact, Management, and Data Resources

I. Executive Summary

The Giant African Snail (GAS) *Lissachatina fulica* is a one of the world's most highly destructive and invasive agricultural pest.[6] Native to East Africa, this mollusk has spread across tropical and subtropical regions globally, including parts of the United States. Its rapid reproductive capacity, broad host range (exceeding 500 plant species), and ability to transmit parasites such as *Angiostrongylus cantonenis* (rat lungworm) make it a significant agricultural, ecological, and public health concern. This review synthesizes the current knowledge on its biology, identification, host range, disease transmission, economic impact, management strategies, with emphasis on peer reviewed literature.

II. Introduction

The GAS is recognized as the most invasive terrestrial invertebrates worldwide. Originating within East Africa, it has expanded to the United States through accidental transportation by cargo and intentional introductions for pets, food and religious practices. Its expansion within society is attributed to hermaphroditic reproduction, high fertility, broad environmental tolerance, and extremely diverse host plant utilization. The GAS feeds on economically important crops, ornamentals, and native vegetation making it a severe threat to agriculture, biodiversity, and infrastructure. They also serve as an intermediate host for zoonotic parasites, elevating public health risks.

III. Biology and Identification of the Giant African Snail (*Lissachatina fulica*)

Life Cycle: Egg, Juvenile (Nymph Equivalent), and Adult Stages

The GAS does not undergo incomplete metamorphosis like insects. Their life stages can be categorized as egg, juvenile, and adult. The Giant African snail reproduces quickly, laying clutches of eggs every 2 to 3 months. It only takes one snail to start a population because they can self-reproduce.[3]

Physical Description and Distinguishing Features

- **Adults:** Adults are usually 3 to 4 inches in length, but some can reach up to 8 inches in length and 5 inches in diameter (about the size of an average adult fist). Adults live for three to seven years, and some individuals can live up to 10 in optimal conditions and can produce 100 to 500 eggs during peak reproductive years.[10][1].
- **Eggs:** Eggs are round, [with] about ¼-inch in diameter, and are yellow to cream colored.[3] They lay their eggs in humid areas with rotting detritus in the top two to six inches of soil.
- **Juveniles:** Young snails may live underground for the first two to six months of their lives.[5] During this time, they develop rapidly and begin feeding once conditions are favorable.

Host Plants and Feeding Behavior

The GAS has a voracious appetite, consuming over 500 different types of plants. They rapidly devour fruits, vegetables, and ornamental plants, wiping out small farms overnight and leaving only foul-smelling excrement in their wake.[5][3] In addition to feeding on living plant tissue, the snail may also damage plaster and stucco structures to obtain calcium needed for shell development.[9]

Geographical Distribution (Global and United States)

The native range of the GAS is in Africa, but it has spread to Asia, the Pacific Islands, the Caribbean, and parts of South America.[6] The original infestation in the 1960s was from a released pet, the current infestation, detected in 2011, was from snails intended for use in religious rituals.[5] In the United States, it was first introduced to Hawaii in 1936 and introduced to the continental U.S. in 1966.[5] The most recent infestations are currently only found in the greater Miami metropolitan area, and one residential area in Broward County; and typically are most active on humid and rainy days, or just after dusk.[5] Eradication programs are coordinated by USDA APHIS in cooperation with state agriculture departments.

Table 1: Key Characteristics of Giant African Snail Life Stages

Life Stage	Size (approx.)	Color/Appearance	Key Distinguishing Features	Typical Location
Eggs	¼ in. in diameter	Yellow-cream	Round, laid in clutches	Humid soil (2-6 inches deep)
Juveniles	Variable	Light brown/cream	Rapid growth, subterranean; Often remain underground for months	Soil surface and underground
Adults	3-8in	Brown, vertically striped shell	Large conical shell; Gray or milky-white body	Soil, vegetation, structures

IV. Disease

The GAS is a threat to both animals and [humans] because it can transmit parasites such as rat lungworm (*Angiostrongylus cantonensis*).[3] In addition, it can carry the gram-negative bacterium, *Aeromonus hydrophila,* causing a wide variety of symptoms, especially in persons with a weak immune system.[7]

Causative Agents

The nematode (roundworm) *Angiostrongylus cantonensis* is a common cause of human eosinophilic meningitis (neural angiostrongyliasis) and occasionally ocular disease.[2] It primarily infects rats but uses snails as intermediate hosts.

Mechanism of Transmission by GAS

Transmission occurs when a rat is infected with the parasite and pass larvae in their feces. The Giant African snail then ingests the droppings while feeding, and the parasite proceeds to develop (molt) within the snail. This parasite can be passed to humans if they consume raw or undercooked infected snails or produce that has been contaminated with snail mucus.[4] In most cases infection does not require medical treatment but, in very rare cases, infection can cause

eosinophilic meningitis, leading to headaches, neck stiffness, and neurological symptoms.[2][4] Most cases resolve without medical treatment, but severe infections may require hospitalization.

Symptoms GASs leave on Plants

- Leaf Symptoms: Irregular holes, defoliation, and slime trails. Severe feeding may strip bark or destroy seedlings.
- Differentiation from Nutritional Deficiencies: Snail damage is irregular and associated with visible feeding scars and slime residue.
- Fruit Symptoms: Limited Information—Research on Fruit symptoms based on the Giant African Snail

Disease Progression and Economic Impact

Infestations can result in significant economic losses due to crop destruction, nursery plant damage, and quarantine restrictions. Past eradication programs in Florida required years of treatment and millions of dollars in funding. The broad host range and long lifespan of the GAS make complete eradication challenging once established.

V. Management and Control Strategies

Integrated Pest Management

Efforts focus primarily on early detection, quarantine enforcement, and coordinated treatment programs.

Monitoring and Early Detection Methods

Residents are encouraged not to handle suspected snails. Instead, they should take a photograph, record the location, and contact state agriculture authorities. Surveillance efforts are coordinated by USDA APHIS and local agencies.

Chemical Control Approaches

Chemical control efforts of the GAS began soon after its introduction into Hawaii in 1936. Between 1938 and 1951, authorities attempted eradication using various pesticide applications.[11] Despite efforts, the population increased rapidly and eradication was eventually considered economically unrealistic because the snails had become widespread. A similar situation occurred in Guam, where the species was first reported in 1945. Control programs used chemical treatments and other methods targeting both rats and snails; however, problems such as limited labor, reduced funding, chemical toxicity concerns in residential areas, lack of spraying equipment, and ineffective barriers allowed infestation to spread across much of the island.[11]

More recent studies have focused on improving chemical control through the use of baiting systems rather than complete eradication. Research found that food-based lures as papaya and bananas increased detection and trapping rates in small areas. While these approaches improved localized mortality, their effectiveness at controlling populations across large regions has not been fully evaluated.[11] Total eradication remains unlikely once populations are established.

Biological Control

After the GAS became abundant in Hawaii, chemical control efforts were considered too costly and ineffective to continue.[11] Agricultural damage was often cited as the primary justification for intervention but public concerns such as the accumulation of snails on structures was also a concern. As a result, an investigation began for biological control options. Early efforts focused on predatory snails from East Africa, including species of *Gonaxis*, as well as the predator *Euglandina rosea* native to Florida. Releases of the *Gonaxis* were conducted in Hawaii and other Pacific islands during 1950s; however, many of these species failed to establish and evidence of their effectiveness in reducing the GAS population was limited or inconclusive. [11]

It was discovered through this control process there were other unintended consequences. The *Euglandina rosea* were highly generalists and preyed on native land snail species, which contributed to declines and extinctions inn Pacific Island ecosystems.[11] Although the GAS populations have declined in some areas, these reductions have not been clearly linked to predator introductions and may be associated with disease, environmental factors, or natural population dynamics. Early biological control programs are now widely regarded as ecologically harmful and largely ineffective at achieving long term suppression of the giant African snail.[11]

Cultural Control Practices

Physical control methods for the Giant African Snail primarily involves the manual removal and habitat management. Handpicking and destroying snails is one of the simplest and most direct control strategies.[11] During the early infestations on Pacific Islands, manual collection was considered effective and relatively inexpensive when populations were still manageable.[11] As snail population dramatically increased, large scale handpicking became impractical due to the labor required and difficulties associated with disposing.[11] However, handpicking remains an effective method in gardens and nurseries where population remains relatively small.

Practices that can continue to reduce the GAS include maintaining clean and well managed growing areas by removing debris, stacked materials, and dense vegetation. These practices reduce humid environments favored by the snails. Keeping lawns mowed, clearing litter around plants, and elevating containers or equipment can limit location for the GAS to hide. Physical barriers may provide some protection; early experiments tested specialized fencing systems, but a simpler approach used bare soil or gravel zones around planting areas to help reduce snail movement into managed areas.[11]

Regulatory Frameworks and Quarantine Programs

Quarantine programs, restrict the movement of soil, plants, and construction materials from infestation zones. Public education campaigns emphasize not buying or releasing live snails and not transporting them from outside the United States.

VI. Available Datasets and Resources

Image Datasets for GAS and Disease Detection

- Collection of images from USDA APHIS and university extension programs with contained labeling of GAS for identification.

Genetic and Genomic Datasets

- Limited Information—Research on this topic for the Giant African Snail is still ongoing or insufficient

Population Monitoring and Disease Incidence Data

- Limited Information—Research on this topic for the Giant African Snail is still ongoing or insufficient

VII. Ongoing Research and Future Prospects

Current research focuses on improving detection methods, understanding parasite to host interactions, and modeling climate suitability for future spread.

VIII. Conclusion

The GAS is one of the most seriously invasive species that threatens agriculture, ecosystems, and public health. Its rapid reproduction, broad host range, and disease transmission capability makes aggressive monitoring and regulatory enforcement necessary. Collaborative efforts between federal agencies such as USDA APHIS, state departments of agriculture, researchers, and the public are critical to prevent re-establishment and spread.

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